Sahaj Bhadja

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Summary

Systems software engineer with 2+ years building **C++ and Python applications**, concurrent processing pipelines, real-time integrations, and computer-vision systems; experienced in object-oriented design, performance analysis, failure isolation, automated testing, and cross-platform delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Engineered **concurrent Python processing pipelines** with background workers, explicit state transitions, and failure isolation, executing 1,000+ jobs while increasing throughput 3x under load and preventing cascading task failures.
- Analyzed and optimized **application performance** across 5,000+ monthly workflows, using validation, reconciliation, indexed access, and production diagnostics to reduce dashboard latency from 1.2 seconds to 350 milliseconds while sustaining 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed a **multithreaded content-processing system** using Python ThreadPoolExecutor and fault-tolerant APIs, securely processing 1,000+ PDF, image, and document inputs per batch while maintaining stable performance under peak load.
- Integrated and tested **computer-vision and OCR pipelines** across multiple model providers, reproducing input-specific failures, refining fallback behavior, and reducing processing turnaround 60% while supporting 10,000+ daily requests.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **Python simulation and evaluation pipelines** for autonomous-system trajectories, combining map geometry and temporal signals to measure performance, compare system behavior, and identify opportunities for reliable feature improvements.
- Analyzed detailed **metrics, constraints, and failure patterns** across experimental runs, communicating reproducible findings to researchers and refining configurations to improve model evaluation quality and accelerate root-cause investigation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed maintainable **object-oriented cross-platform software** using Singleton and Factory patterns, API integrations, SQL, and caching, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and overall maintainability.
- Automated versioned builds, performance validation, and **20+ unit and integration tests** across Windows, Linux, and macOS, collaborating through code reviews to diagnose defects and deliver reliable production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Implemented **real-time WebSocket and REST integrations** across eight services and 2M+ synchronized records, reducing cross-location conflicts 33% while maintaining responsive updates for resource, billing, permissions, and notification features.
- Monitored and optimized feature performance using Sentry, Azure Monitor, Kibana, and automated browser tests, isolating production failures and shipping verified Docker releases through Jenkins while reducing deployment time 50%.

Skills

Languages: C++, C, Python, Java, SQL, Bash, JavaScript, TypeScript
Systems: OOP, data structures, algorithms, multithreading, concurrency, synchronization, CPU scheduling, memory, I/O
Media & Vision: PyTorch, Transformers, computer vision, OCR, image processing, real-time pipelines, performance analysis
Networking: TCP/IP, WebSockets, REST APIs, client-server architecture, distributed systems, low-latency optimization
Quality & Platform: Linux, Windows, Git, Jenkins, GitHub Actions, CI/CD, Docker, PyTest, Mockito, debugging, documentation

Projects

CPU Scheduling Simulator Live | *C++, Algorithms, Systems Performance*
- Implemented and validated **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case handling, and live metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.

CaptionCraft Live | *PyTorch, Transformers, Flask, Computer Vision*
- Benchmarked **four transformer vision architectures** across 10,000 images and 100,000 captions, improving accuracy from 81% to 96% while building scalable Flask inference APIs for real-time image caption generation.

PiXMod Live | *PyTorch, SAM, Stable Diffusion, MongoDB*
- Built an **image-processing and segmentation pipeline** using SAM, Stable Diffusion, and PyTorch, processing 1,000+ edited assets while storing images and metadata in MongoDB for scalable retrieval, diagnostics, and analysis.